

May 2024

(BT)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. **HeLa cells are derived from cervical cancer from _____**
A) Henry Lacks
B) Henrietta Lacks
C) Henrietta Louis
D) Henry Lapse
2. **Which type of cytoskeletal filament is responsible for cell movement and cell shape changes?**
A) Microtubules
B) Intermediate filaments
C) Microfilaments
D) All of the above
3. **Which of the following is not a type of cell membrane transport?**
A) Diffusion
B) Osmosis
C) Filtration
D) Secretion
4. **Which of the following is NOT a function of mitochondria?**
A) Production of energy
B) Cell division
C) Regulation of cell death
D) Calcium storage
5. **Which of the following is not a component of the cell membrane?**
A) Lipids
B) Proteins
C) Nucleic acids
D) Carbohydrates
6. **Which disease is associated with a malfunctioning endoplasmic reticulum?**
A) Alzheimer's disease
B) Cystic fibrosis
C) Huntington's disease
D) All of the these
7. **In which phase of the cell cycle do the centrosomes duplicate?**
A) G1
B) S
C) G2
D) M
8. **What is the resolution limit of transmission electron microscopy (TEM)?**
A) 0.2 nm
B) 2 nm
C) 20 nm
D) 200 nm
9. **Which protein is responsible for generating force during muscle contraction?**
A) Actin
B) Myosin
C) Tubulin
D) Keratin
10. **Which protein is responsible for anchoring cells to the extracellular matrix in focal adhesions?**
A) Cadherin
B) Selectin
C) Integrin
D) Actin
11. **What is the function of second messengers in signaling pathways?**
A) To amplify the signal
B) To terminate the signal
C) To transport the signal across the plasma membrane
D) To sequester the signal within the cell
12. **Which type of signaling molecule is released into the bloodstream to act on distant target cells?**
A) Autocrine
B) Paracrine
C) Endocrine
D) Synaptic

13. Which of the following is a downstream effector protein of tyrosine kinase signaling?
 A) Adenylate cyclase B) Phospholipase C
 C) Protein kinase A D) Ras
14. Which of the following is an example of a food product that is produced through fermentation?
 A) Yogurt B) Bread
 C) Beer D) All of the these
15. Which of the following is a limitation of using microbial indicators in environmental monitoring?
 A) They are not sensitive to all types of environmental pollutants B) They can produce false-positive or false-negative results.
 C) They require a large amount of equipment and technical expertise. D) All of the above
16. Who discovered 'animalecules'
 A) Anton van Leuwenhook B) Rudolf Virchow
 C) Kary Mullis D) Theodore Schwann
17. Who is known as Father of Microbiology?
 A) Louis Pasteur B) Edward Jenner
 C) Robert Hooke D) Karl Erkey
18. The principle of _____ microscope is 'Light is transmitted through the sample and absorbed by it'.
 A) Phase contrast B) Bright field
 C) Fluorescent D) Dark field
19. _____ was first observed in axoplasm of myelinated nerve fiber by Robertis and Franchi
 A) Microfilament B) Intermediate filament
 C) Microtubule D) Actin filament
20. Which of the following cell adhesion receptor is involved in heterophilic adhesion
 A) Cadherin B) Mucin
 C) Ig superfamily D) Actin

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) How does the Cell Theory explain the organization and structure of multicellular organisms? Provide examples to support your answer.	08	L*1	1	1,12
	b) Discuss the role of organelles in cell diversity. Provide examples of organelles that are unique to specific cell types and describe their functions.	08	L1	1	1,12
2.	a) Describe the roles of lysosomes in cellular digestion and waste disposal. How do these organelles contribute to the maintenance of cell health and homeostasis?	08	L1	1	1,12
	b) What are the key events that occur during Meiosis-I? How does meiosis contribute to genetic variation?	08	L1	2	4,12
3.	a) Describe the main components of a compound light microscope with the help of a neat, labelled sketch.	08	L1	2	4,12
	b) Give a contrast note on: i) Live cell imaging and fixed cell imaging, ii) Transmission Electron Microscopy & Scanning Electron Microscopy.	08	L1	2	4,12
Unit – II					
4.	a) Narrate the functions of cell junctions with pictorial representation.	08	L1	3	1,12
	b) Discuss about the general principles of cell signaling process and add a note on different types of signaling molecules.	08	L1	4	1,12

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| 5. | a) | Write a comparative note on Cadherin and Ig family cell adhesion molecules. | 08 | L1 | 3 | 1,12 |
| | b) | Cells receive and process a diverse set of chemical signals and sensory stimuli, Elaborate. | 08 | L1 | 4 | 1,12 |
| 6. | a) | Write the significance of G-protein coupled receptor in eukaryotic signaling pathway. | 08 | L1 | 4 | 1,12 |
| | b) | Give a contrast note on Microtubule and Actin filament. | 08 | L1 | 3 | 1,12 |

Unit – III

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|----|----|---|----|----|---|------|
| 7. | a) | Microbial cells play a role in medical diagnosis. Justify with suitable examples. | 08 | L1 | 5 | 1,12 |
| | b) | Comment on microbial biosensors for water and soil analysis. | 08 | L1 | 5 | 1,12 |
| 8. | a) | Discuss different types of stem cells and add a note on stem cell therapy. | 08 | L1 | 5 | 1,12 |
| | b) | Write the pros and cons of cellular modifications using modern tools. | 08 | L1 | 5 | 1,12 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
